Regularization (L1 & L2) — The Art of Taming Overly Smart Models

Can a model be "too good' at learning? Turns—yes, osss, It's ta—when it memorizes the training data so perfectly that t' can't generalize to new situations.

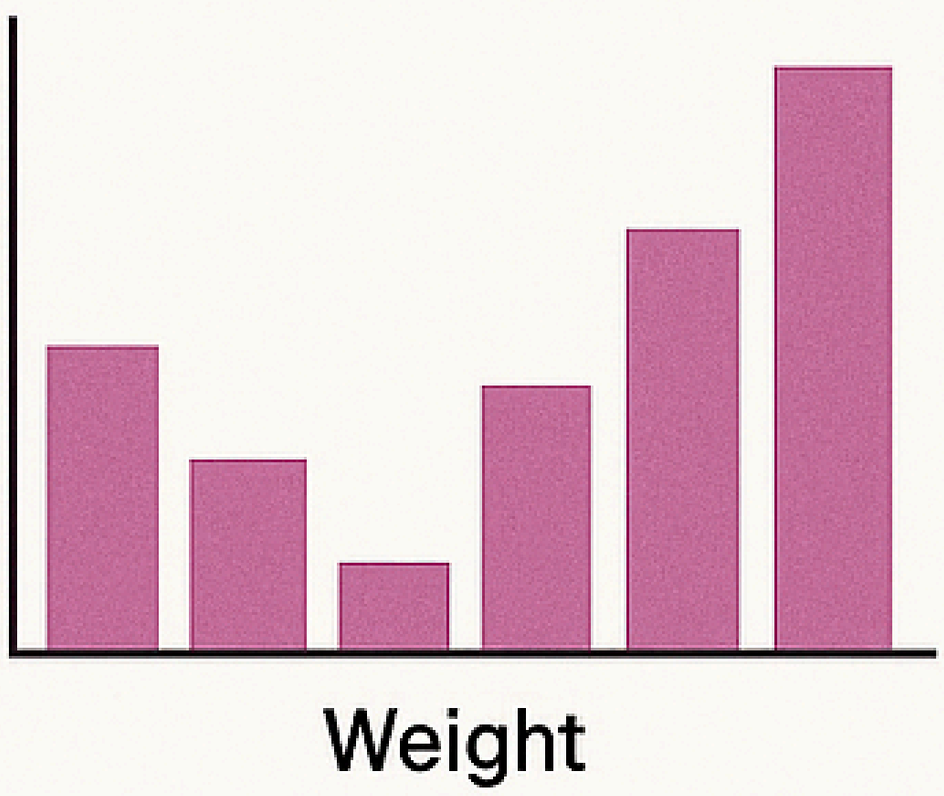
Regularization is a technique that adds a penaly taying *pulles* a penalty to the model when it becomes to complex. It basically says a smaller weights='minpler model = better generalization.

The goal? to prevent the model from relying too much on individual features and instead focus on the overall patterns in the data.

How Does It Work?

Adding a penality term' to Loss function. Which the model tries to minimize while learning.

- L1 Regularization (Lasso)
- L2 Regularization (Ridge)



Why use L1?

- It can 'force some weights to become exactly 0. that means it actually performs a feature selection.
- Useful when you'v have lots of features and want to ignore irrelevant ones